Object Detection in High-Resolution Remote Sensing Images Based on a Hard-Example-Mining Network

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***Abstract*— In recent years, object detection in remote sensing images (RSIs) has attracted much attention for its application value. Compared with traditional methods that are based on manually extracted features, deep learning methods have a great advantage for object detection and have been vastly promoted. However, existing deep learning methods leave much to be desired in the field of RSI object detection due to the large-scale range of the objects and the complex image backgrounds in RSIs. Algorithms need to be specially optimized for this situation. To solve this problem, we propose an effective deep learning-based RSI object detection framework called the multiscale hard-example-mining network (MSHEMN), which is composed of three parts. First, we use the existing ResNet-50 for feature extraction. Second, we propose a multiscale region proposal network (MSRPN), which improves the existing top–down pathway feature pyramid architecture of feature pyramid network (FPN) by adding lateral connection block (LCB) and adaptive feature merge (AFM) to extract features that combine high-resolution and strong semantical information. Finally, a hard-example-mining network (HEMN), which is a cascade multistage detection network integrated with a hard example mining strategy, is proposed to make the detection network focus on hard examples during the training phase by changing the input data distribution of each stage. Extensive experiments on the High-Resolution Remote Sensing Detection (HRRSD) data set have shown the effectiveness of our proposed method, which achieves an average precision (AP) of 62.6 on the testing data set.**

***Index Terms*— Convolutional neural networks (CNNs), hard-example-mining network (HEMN), multiscale region proposal network (MSRPN), object detection, remote sensing images (RSIs).**

I. Introduction

AEROSPACE remote sensing technology has developed for decades, and the number and quality of remote sensing images (RSIs) have increased dramatically. A large number of RSIs have been utilized for multiple purposes, such as agriculture, city planning, and geological exploration. Object detection is a fundamental problem in RSI analysis that has attracted considerable attention from researchers and has been extensively studied.

Over the past years, many remarkable advances have been made in the task of object detection in RSIs [1]–[9]. Object detection contains two objectives: localization and recognition; the former is a regression problem, and the latter is a classification problem. In general, object detection is performed in feature space, so it is important to design an effective feature extractor. According to the chosen feature extractors, the method of object detection in RSIs can be divided into two main classes: traditional manual methods and deep learning-based methods.

The traditional manual methods extract features manually for object detection in RSIs. Many of these features are low-level features. However, these features are not competent to perform the task of object detection in images with complex backgrounds and noise. Furthermore, it is a difficult task to manually design a robust and efficient feature for object detection in RSIs; a great deal of particular knowledge of RSIs is needed, and tremendous parameters need to be tuned carefully.

In recent years, deep learning-based object detection algorithms have made great progress [10]–[18]. Deep learning-based methods use convolutional neural network (CNN), which contains several cascaded convolution, ReLu, and pooling layers to extract features from images. Finally, the detection network uses the features extracted by CNN to identify the class of the object and tune the bounding box to obtain the final location and size of the object. Compared with the traditional manual methods, deep learning-based methods can extract more robust and high-level features from raw pixels without any extra processes. Though deep learning-based object detection algorithms have achieved great breakthroughs in general images, however, the detection result leaves much to be desired for the RSIs [19], and the main reason is that in contrast to general images, RSIs have several special features that make detection more difficult.

First, objects in RSIs have a huge size imbalance, and the length of the short side of small objects, such as cars and yachts, is only approximately 10 pixels, whereas the length of the long side of some large objects, such as ground track fields and large-scale bridges, is over hundreds of pixels, which requires the detection algorithm to deal with objects in a large range of scales.

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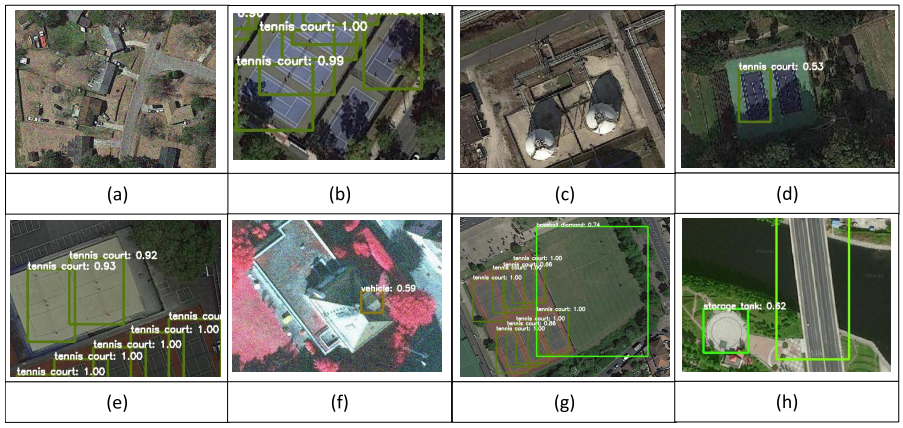


Fig. 1. Examples of detection errors caused by complex background in RSIs. Faster R-CNN with FPN is used as detection network. In (a) and (b), the detection error is caused by occlusion; in (c) and (d), shadow causes the detection error; and in (e)–(h), similar color and shape objects in background cause detection error.

Second, RSIs usually have a large field of view and do not have a particular target; therefore, the background of RSIs is more complex and diverse than that of general images. The background of RSIs contains both natural and artificial elements, such as vegetation, bare soil, naked rock, ocean, and buildings. These complicated backgrounds will substantially disturb object detection. This is another great challenge to object detection. The shadow area also brings interference, especially when there is sun slant. What is more, objects in RSIs are not uniformly distributed. A single large RSI may contain just few objects, while the rest of the image is background, which increases the imbalance between positive examples and negative examples. In the meanwhile, quite a large part of the background in RSIs is simple and can be classified easily, such as bare soil, which makes the imbalance between easy examples and hard examples more serious.

Existing deep learning-based algorithms are not specially optimized for these problems. Due to the existence of pooling layers, the size of the features that extracted by CNN is much smaller compared with the size of the original image. The resolution of features is low and fixed, which makes it difficult for the network to obtain the exact location of multiscale objects in RSIs, especially those small objects. Existing methods assign the same weight to all of the examples, and during training, the easy examples and negative examples dominate the total loss, which causes inefficient training process. Therefore, the detection network cannot focus on objects with complex backgrounds.

Take Faster R-CNN with feature pyramid network (FPN) for example. In Fig. 1, we list some typical detection errors caused by the complex background in RSIs. As part of the object is overshadowed by trees, the T-junction in Fig. 1(a) and the tennis court in Fig. 1(b) cannot be detected. The shadow of the surrounding causes missed detection in Fig. 1(c) and (d). As shown in Fig. 1 (e)–(h), the color, shape, and texture of some objects that belong to the background are similar to the targets, so the detectors may confuse them with targets.

To solve the aforementioned problems in object detection in RSIs, we propose a dedicated detection framework called multiscale hard-example-mining network (MSHEMN), which contains three main parts: CNN feature extraction, multiscale region proposal network (MSRPN), and HEMN. Considering the tradeoff between speed and performance, we use ResNet-50 [20] for feature extraction. In MSRPN, we improve the top–down pathway feature pyramid architecture of FPN [21] in two aspects. First, the lateral connection block (LCB) is used as the lateral connection to obtain a better representation of the bottom feature. Second, we use adaptive feature merge (AFM) to merge all the feature maps of different scales to obtain a feature that combines high-resolution and strong semantical information. In HEMN, we integrate a hard example mining strategy with a cascade multistage detection network so that the network can focus on hard examples during the training phase to reduce the detection mistakes caused by the complex background in RSIs. More concretely, during training, regressors of the former stage produced C refined bounding boxes with the corresponding scores and labels (C is the number of total classes) from one proposal. After that, these bounding boxes are ranked according to the score and the remaining top K bounding boxes decide the proposal region of the next stage. What is more, an adaptive weight is multiplied to the classification loss function according to the score and label of the proposal to decrease the influence of easy example on model update and increase the influence of hard example, so the detector of the latter stage will focus on the hard examples and the network will be trained more efficiently.

To evaluate our proposed model, we conduct extensive experiments on the High-Resolution Remote Sensing Detection (HRRSD) [22] data set. The results show that our model achieves good performance on the data set and outperforms some state-of-the-art object detection approach. The main contributions of this article are summarized as follows.

1) We propose a CNN-based network named MSHEMN to improve the object detection performance on RSIs. MSHEMN can achieve a good performance in detecting objects that have a large-scale range. What is more, it can reduce the imbalance between easy examples and hard examples in RSIs and address the detection error caused by the complex background.

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2) We proposed MSRPN, which is an effective multiscale region proposal generator and feature extraction network that can obtain multiscale features and merge them together to obtain features that combine high-resolution and strong semantical information. The framework improves the multiscale object detection accuracy and achieves good performance on objects in RSIs.
3) We also proposed a cascade multistage detection framework with the hard-example-mining mechanism called HEMN. During training, the multistage detectors are trained sequentially, and those negative examples that can be classified easily are removed before sent to the next stage. We also add some positive examples according to the former stage detection result to help training. Furthermore, we give relatively greater weights to hard examples. By doing this, we change the distribution of the input training data of the latter detector, which equivalently forces the latter detector to focus on the hard examples. By using this architecture, we successfully reduce the detection errors caused by complicated background on RSIs.

The rest of this article is organized as follows. Section II introduces existing methods of object detection in RSIs. Section III describes the proposed framework of detection in detail. In Section IV, implementation details are illustrated, and experimental results are reported and discussed. Finally, conclusions are made in Section V.

II. Related Work

A. Deep Learning-Based Object Detection Framework

A large number of studies have been performed for RSI object detection. These studies can be divided into two categories according to the feature extractor: traditional manual methods and deep learning-based methods. Traditional manual methods use handcrafted features to detect objects. Local binary pattern (LBP) [23], [24], scale-invariant feature transform (SIFT) [6], [25], histogram of oriented gradients (HOG) [1], [26], and bag of words (BoW) [27] are the most frequently used handcrafted features in RSI object detection.

However, it is a challenging task to design a robust and effective feature to detect objects, especially in complex situations. In recent years, many outstanding CNN models were proposed [20], [28]–[31], which promotes the development of deep learning algorithms in a wide range of computer vision tasks; for instance, face detection [32], coherent groups detection [33], and urban scenes semantic segmentation [34]. Compared with the traditional manual methods, deep learning-based methods can learn to extract high-level features from images without human involvement, designing a suitable CNN to learn how to extract features replaces designing representation features directly. The features extracted by CNN are also more powerful and robust, which leads to a significant improvement in detection performance. Due to its convenience and efficiency, object detection that uses CNN to extract features has received increasing attention.

There are two basic schemes: a two-stage scheme and a one-stage scheme. The framework of two-stage detection networks can be mainly divided up into two parts: one part generates class-agnostic region proposals, and the other part infers the class and coordinate of each proposal. Spatial pyramid pooling net (SPP-Net) [35], region-based CNN (R-CNN) [10], Fast R-CNN [11], Faster R-CNN [12], Region-based fully convolutional network (R-FCN) [13], and Mask R-CNN [14] are the typical methods of two-stage detection networks. Take Faster R-CNN for example, and it consists of a framework containing four main components: the CNN, the region proposal network (RPN), region-of-interest (RoI) pooling, and detection network. CNN is used to extract features from images. RPN is used to generate region proposals based on anchors. It uses SoftMax to estimate whether an anchor belongs to the foreground or background and tunes the bounding box of the anchor to achieve a more accurate proposal location. RoI pooling obtains features corresponding to the region proposals and converts them to a fixed-length feature vector. Finally, the detection network uses the features of the proposals to infer the class of objects and get precise coordinate. Both RPN and the detection network share the features extracted by CNN. One-stage detection networks replace region proposals by anchor box, and the class and coordinate of objects are directly identified via a single class-specific detection. You Only Look Once (YOLO) [15] was the first proposed one-stage detection method, and single-shot multibox detector (SSD) [16], YOLOv2 [17], YOLOv3 [18], deconvolutional single-shot detector (DSSD) [36], deeply supervised object detector (DSOD) [37], and RetinaNet [38] are the typical methods of one-stage detection networks.

Compared with one-stage detection networks, two-stage detection networks are slower but have higher accuracy, and therefore, in the remote sensing field, most of the deep learning object detection methods use two-stage detection networks. Long *et al.* [4] and Cheng *et al.* [5] used selective search algorithms to achieve region proposals and used CNN to extract features. Specifically, Long *et al.* [4] used AlexNet [29] and GoogleNet [31] as models to extract features and proposed an unsupervised score-based bounding box regression (USB-BBR) algorithm to improve the location accuracy. In [5], a rotation-invariant CNN (RICNN) model is introduced by implementing a rotation-invariant layer based on the AlexNet CNN model. Zhang *et al.* [22] added a rotation and scaling robust enhancement (RSRE) structure to the CNN model to deal with various factors of rotation and scaling simultaneously. Lu *et al.* [39] proposed a model called gated axis-concentrated localization network (GACL Net) to improve the performance of small-scale object detections in RSIs that used global features to guide the channel attention of local features and avoided coordinate prediction disturbance by using a single-axis pooling process.

B. Multiscale and Multistage Object Detection

The large-scale range of objects is one of the most important problems in object detection, especially in RSIs. A large number of studies have attempted to deal with this problem. Image pyramid [32], [40] is a widely used scheme to handle scale variation by resizing images to a set of different

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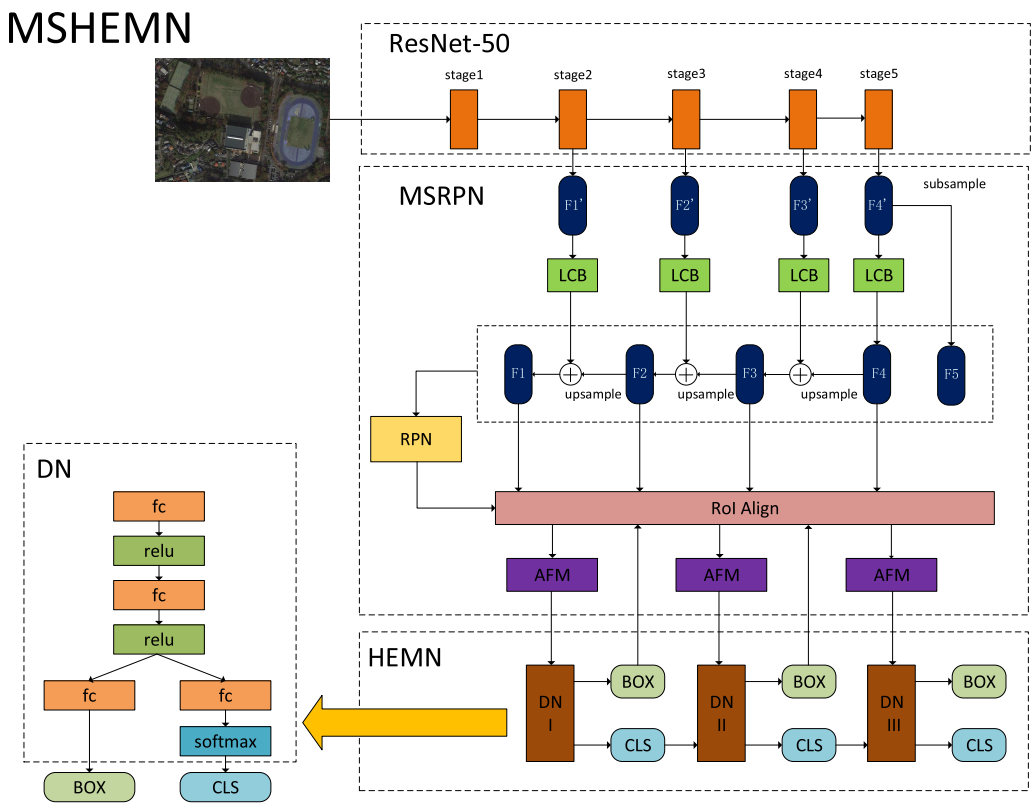


Fig. 2. Architecture of the proposed MSHEMN. It consists of a ResNet-50 feature extraction network, an MSRPN, and an HEMN. The BOX represents the position of the object, the CLS represents the probability of the category, the fc represents fully connected layer, and the relu represents the activation function.

scales before sending them to the network. Image pyramid is effective but has a relatively high computation cost. Singh and Davis [41] proposed a new method called SNIP that trains and tests detectors on the same scale by selectively backpropagating the gradients of objects of different sizes as a function of the image scale. Instead of using multiple images of different sizes, another method uses multilevel features to solve this problem. SSD [16] combined predictions from multiple feature maps with different resolutions to naturally handle objects of various sizes. FPN proposed a top–down pathway feature pyramid architecture with lateral connections to extract semantic feature maps at all scales. Compared to FPN, path aggregation network (PANet) [42] added an additional bottom–up path to enhance the feature representation and used adaptive feature pooling to aggregate the features of different layers, in order to have a better prediction result. RefineDet [43] combined SSD with FPN and introduced transfer connection block (TCB) to convert features of different layers from the anchor refinement module (ARM), and TCB integrated large-scale context by adding high-level features to the transferred features to improve detection accuracy. To detect vehicles in RSI, Gao *et al.* [44] proposed the MSRPN, which modified the feature pyramid architecture of FPN. Features from top level and bottom level were excluded and proposals were mapped to the finest stage to fit in with the needs of vehicles' detection.

Some works [40], [45]–[48] have argued that a single regression step in the second stage of Faster R-CNN is not sufficient for object detection. To solve this problem, Gidaris and Komodakis [48] proposed a powerful localization system that used an iterative CNN-model, which alternated between scoring candidate boxes and refining their coordinates for bounding box regression. Zagoruyko *et al.* [45] proposed integral loss, which was a loss function that optimized an integral of localization overlaps to improve localization. Cai and Vasconcelos [46] proposed a multistage object detection architecture called Cascade R-CNN, which consisted of a sequence of detectors, trained with increasing intersection-over-union (IoU) thresholds.

III. Proposed Framework

To deal with the aforementioned problems, we propose a dedicated framework called MSHEMN to detect objects in RSIs. In this section, we will discuss the details of this framework. Our framework is shown in Fig. 2, and it is divided into three main parts: CNN feature extraction, MSRPN, and HEMN. First, we apply ResNet-50 [20] to extract features. Second, we use the MSRPN to fuse the features from different scales, intending to obtain scale-robust features and the regions that may contain the objects. Finally, we use a multistage detection architecture called HEMN to analyze the features of the proposals, identify the class of object, and refine the bounding box to decide the final location and size of the object.

A. Multiscale Region Proposal Network

Many deep learning-based methods only extract single-scale features to detect an object; for example, Faster R-CNN [12]

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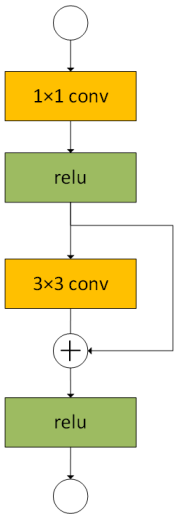


Fig. 3. Architecture of the proposed LCB. "conv" represents the convolutional layer, and "relu" represents the activation function.

uses a feature map that is 1/32 the size of the input image, which is not suitable for the large-scale range of objects, especially those small objects in RSIs. To solve this problem, we propose the MSRPN to extract multiscale features from images. The MSRPN is based on FPN, and its architecture is shown in the middle part of Fig. 2. We use ResNet-50 [20] as the backbone, which has five-stage convolutional networks. We obtain four different scales of feature maps F'_1–F'_4 from the last residual block of the second stage to the fifth stage. The four scales correspond to 1/4, 1/8, 1/16, and 1/32 of the input image size. Thus, we have higher resolution feature maps for those small objects and can obtain more useful information.

To obtain a more efficient bottom-level feature, we merge the top-level feature and bottom-level feature via top–down pathways and lateral connections

$$F_n = L\left(F'_n\right) + U(F_{n+1}), \quad (n = 1, 2, 3) \tag{1}$$

$$F_n = L\left(F'_n\right), \quad (n = 4). \tag{2}$$

$U(.)$ indicates upsample because the resolution of top-level feature is lower than that of the bottom-level feature, and the top–down pathway upsamples the top feature map by a factor of 2 before merging. $L(.)$ is the lateral connection and differs from the original FPN, which uses a 1×1 convolution as the lateral connection to reduce the number of feature maps in the bottom-level feature. The reduction of the number of channels brings loss to the feature maps, so we use an LCB to further refine the feature maps and obtain a better representation of the bottom feature. The proposed architecture of LCB is shown in Fig. 3. We use a convolution layer with a 1×1 kernel to adjust the channel of the input feature map to 256, followed by a residual block that uses a 3×3 convolution layer as a residual function to refine the feature maps.

We also subsample F'_4 by a factor of 2 to obtain F_5, in which scale is corresponding to 1/64 of the original image size, to adopt the objects that have a large scale, such as ground track fields and large-scale bridges.

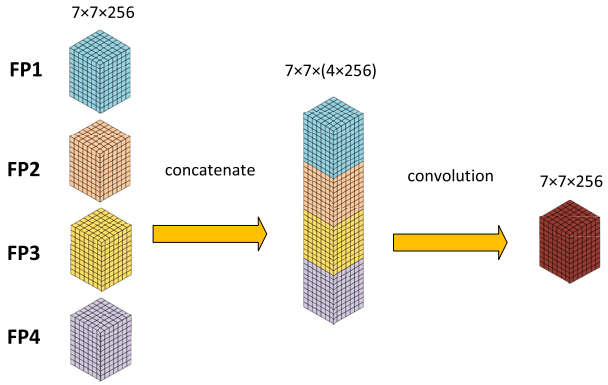


Fig. 4. AFM structure: features of different scales are concatenated and then merged by a set of $1 \times 1 \times 1024$ filters.

Top-level features are acquired through several convolution layers and max-pooling layers, so they have more semantical information but lower resolution. On the other hand, bottom-level features have higher resolution, which means that they have more locational information but are semantically weak. To obtain a feature that combines high-resolution and strong semantical information, we propose using AFM to merge four scale feature maps F_1–F_4. The structure of AFM is shown in Fig. 4. Original FPN assigns an RoI to a specific feature map level according to the width and height of the RoI and then implements RoI pooling on this feature map. Unlike the abovementioned process, we use an RoI align layer, which is proposed in [14], to divide the feature maps on the corresponding RoI position of four feature maps F_1–F_4 into 7×7 fixed-length feature maps FP_1–FP_4 separately. Then, we concatenate these feature maps together to form a 7×7 feature map that has 1024 channels. After that, we let the network learn how to combine features across multiple scales by using a set of $1 \times 1 \times 1024$ filters to transform the feature map into 256 channels. It should be noted that as the RoI align layer transforms feature maps of different scales to the same size, additional upsampling or downsampling processes on the feature maps are not needed.

B. Hard-Example-Mining Framework

Traditional deep learning-based methods have unsatisfactory detection results in RSIs with complex backgrounds. This is partly because objects cannot match enough samples during training, which leads to the imbalance between positive examples and negative examples. The second reason is the imbalance between easy examples and hard examples. For a particular sample, if it can be classified by the model easily, we call it an easy example. The network cannot learn much useful information from these samples during training. In contrast, some samples are difficult for the model to classify, and we call these hard examples. These samples can give more information to the model for optimization during training. During training, easy examples occupy a large proportion of the proposals extracted by the RPN, which causes the easy examples to dominate the total loss. However, these examples cannot update the parameters of model effectively, which makes the training process inefficient.

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TABLE I
DISTRIBUTION OF NUMBER OF TARGETS IN HRRSD

number of targets	1	2	3	4	5	6~10	11~20	>20
number of images	9446	4911	2791	1686	1005	1453	427	42

To solve this problem, we extend the detection subnetwork to a multistage framework. Our idea is inspired by Cascade R-CNN [46]. The architecture of HEMN is shown in the bottom of Fig. 2; it is a cascade multistage detection network, and the proposal region of the next stage is decided by the bounding boxes produced by the former stage. The features are reextracted according to the new proposals before being sent to the detection network.

During training, the multistage detectors are trained sequentially; the latter stage receives N proposals from the former stage and the output is $\mathrm{N} \times \mathrm{C}$ refined bounding boxes with the corresponding scores (C is the number of total classes). Then, before being sent to the next stage, nonmaximum suppression (NMS) is applied to these bounding boxes. After that, the remaining bounding boxes are ranked according to the score. We only retain top K bounding boxes to reduce redundancy and remove the proposals that are obviously background. Objects in RSIs are not uniformly distributed. Take the HRRSD data set for example, and we calculate the distribution of the number of targets in HRRSD. As shown in Table I, about 91% of the images contain less than six targets, more than 8% of the images contain 6–20 targets, and only about 0.2% of the images contain more than 20 targets.

Applying a large K on those images that contain just a few of targets will increase the imbalance between positive examples and negative examples, and a small K will reduce the detection performance on those images that contain a large number of targets. To overcome this contradiction, in the training phase, we make K adaptive according to the number of targets that can be obtained from ground truth. K value is calculated as follows:

$$K = \begin{cases} \dfrac{N_{\mathrm{Max}}}{4} + (N_{\mathrm{Tar}} - 1) \times \dfrac{N_{\mathrm{Max}}}{4 \times (N_1 - 1)} & N_{\mathrm{Tar}} \leqslant N_1 \\ \dfrac{N_{\mathrm{Max}}}{2} + (N_{\mathrm{Tar}} - N_1) \times \dfrac{N_{\mathrm{Max}}}{2 \times (N_2 - N_1)} & N_1 < N_{\mathrm{Tar}} \leqslant N_2 \\ N_{\mathrm{Max}} & N_2 < N_{\mathrm{Tar}} \end{cases} \tag{3}$$

where N_{Tar} is the number of targets and N_{Max} is the maximum value of bounding boxes to be reserved, and N_1 and N_2 are the thresholds of the number of targets that are determined according to the data sets. In HRRSD, we set N_1 as 5 and N_2 as 20. In the testing phase, as the number of targets cannot be obtained in advance, we just ignore the proposals with score less than 0.001 for speeding up. After filtering the bounding boxes, the remaining bounding boxes are used as the proposals for the next stage, in which they are defined as positive or negative examples according to the IoU with ground truth. The IoU thresholds of three detectors are set as 0.5, 0.6, and 0.7 separately to reject close false positives (FPs).

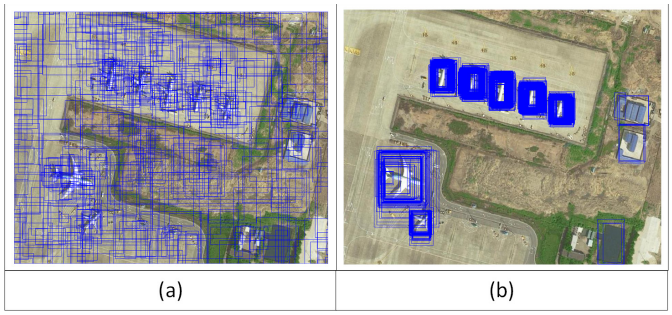


Fig. 5. Proposals produced by (a) RPN and (b) second stage.

In Fig. 5, we show the comparison result between proposals produced by RPN and proposals produced by the second stage in our method. We can see that the proposals produced by the second stage eliminate most of the background that can be easily classified and many proposals are localized around the objects. Hence, we can obtain adequate positive samples in the last stage.

The multitask loss function defined in [11] is

$$L_{p,y,u,v} = L_{\mathrm{cls}}(p, y) + \lambda[y \geq 1] L_{\mathrm{loc}}(u, v) \tag{4}$$

where p and y are predictions and labels of proposals, respectively, u and v represent the bounding boxes regression results and corresponding ground truth, respectively, and λ is the tradeoff coefficient.

The classifier in Faster R-CNN is a function $f(x)$ that uses the feature maps of proposals as input and outputs a $\mathrm{C} + 1$ dimension array that represents the probability of a proposal belonging to $\mathrm{C} + 1$ classes $f_i(x) = p(y = i|x)$, where an additional class represents the background. It uses cross entropy as a loss function of the classification

$$L_{\mathrm{cls}} = -\sum_{j=1}^{N} y_j \log p_j. \tag{5}$$

During the training phase, given a training example with input x and label $y = i$, the classification loss is

$$L_{\mathrm{cls}} = -\log f_i(x). \tag{6}$$

To make the detection network focus on the hard examples, we multiply an additional adaptive weight S^{γ} in the original cross-entropy loss function of the detection network, except for the first stage

$$L_{\mathrm{cls}} = -S^{\gamma} \log f_i(x) \tag{7}$$

where S is the detection accuracy of the suspected object that is output from the former stage and γ is a positive factor that determines how much the former stage affects the next stage. After the bounding boxes are assigned as a positive or negative example according to the ground truth, S is determined by the label and score of the bounding box. The detailed expression is as follows:

$$S = \begin{cases} P & \text{If bbox is a negative example} \\ 1 - P & \text{If bbox is a positive example} \end{cases} \tag{8}$$

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where P is the score of the bounding box, in which the former stage detector infers. If a bounding box is regarded as a positive example, the higher score indicates the better performance that the former detector has on this object. When P is close to 1, which means that this object is relatively simple to classify, S will be small and will reduce the corresponding L_{cls}. Therefore, the next stage detector will pay less attention to this sample. In contrast, if P is small, which means that this object is difficult to classify, L_{cls} will be large. In this situation, the next stage detector will pay more attention to this object. If a bounding box is regarded as a negative example, a large P indicates that it is a hard example. The corresponding L_{cls} will increase and thus make the next stage detector focus on this object.

Balanced L1 [49] is used in L_{loc}, and the detailed functions are listed as follows:

$$\text{BalancedL1(x)} = \begin{cases} \frac{\alpha}{\beta}(\beta|x|+1)\ln(\beta|x|+1)-\alpha|x| & \text{if}|x|<1 \\ \gamma|x|+C & \text{otherwise} \end{cases} \tag{9}$$

and the parameters γ, α, and β have to satisfy the following conditions:

$$\alpha\ln(\beta+1)=\gamma. \tag{10}$$

We set γ as 1.5 and α as 0.4, 0.3, and 0.2 for three-stage detectors.

Our method is different from Cascade R-CNN [46], which is a multistage extension of the R-CNN with a resampling mechanism applied on. In Cascade R-CNN, cascade regression is used as resampling mechanism. All bounding boxes are class agnostic and resampling is implemented by using the regressed outputs from the previous stage and outliers are sequentially removed by increasing IoU thresholds to obtain a better training result. The resampling mechanism of Cascade R-CNN does not aim to mine hard negatives, and it tries to find a set of positives with higher IoU for training the latter stage by adjusting bounding boxes.

In HEMN, during training, one proposal produced C refined bounding boxes with the corresponding scores and labels (C is the number of total classes) by regressors. These bounding boxes are ranked according to the score and only top K bounding boxes are sent to the next stage. Because we want to produce more positive examples during training for the next stage to learn, we can balance the number of positive examples and negative examples. Also, although some hard positive examples are given a small score in the former detecting stage, their scores are still significantly higher than those examples that can be easily classified as background. By picking up the top K bounding boxes according to the score as the proposals of next stage, we remove those proposals that are obviously background, thus reducing both the number of negative samples and easy examples at the same time.

An additional adaptive weight is also multiplied to the classification loss function according to the score and label of the proposal, which does not exist in Cascade R-CNN. By adding this weight to the loss function, we decrease the influence of easy example on the model update and increase the influence of hard example, thus relieving the imbalance of easy and hard examples. This is equivalent to changing the distribution of input data of the latter detector. Therefore, the latter detector will focus on the hard examples and update the model more efficiently.

IV. Experiments

A. Data Set Description

We evaluated our proposed method on the HRRSD [22] data set, which includes 21 761 images. These images are acquired from Google Earth, and the spatial resolution is spread from 0.15 to 1.2 m. Among these images, 5401 images are assigned to train subset, 5417 images are assigned to val subset, and 10 943 images are assigned to test subset. HRRSD contains 13 categories: airplane, baseball diamond, basketball court, bridge, crossroad, ground track field, harbor, parking lot, ship, storage tank, T-junction, tennis court, and vehicle. Each category has approximately 4000 annotated objects, each of which is labeled by an axis-aligned quadrilateral. Compared with existing RSI object detection data sets such as NWPU VHR-10 [5] and UCAS-AOD [50], HRRSD is much larger and has a richer sample. Furthermore, HRRSD is a category-balanced data set, that is to say, for each category, the ratio of the number of instances in the train, val, and test subsets is approximately 1:1:2. Therefore, we can use HRRSD to verify the effectiveness and robustness of our proposed framework.

B. Evaluation Metrics

Average precision (AP) is widely used in the evaluation of object detection and is treated as a standard evaluation indicator. In our experiments, we adopted this metric to evaluate our framework.

To judge whether the detection bounding box corresponds to the ground truth, we used IoU as an indicator. IoU can be formulated as follows:

$$\text{IoU}=\frac{\text{GT}\bigcap\text{BD}}{\text{GT}\bigcup\text{BD}} \tag{11}$$

where GT represents the ground truth of the image and BD represents the bounding box of the detection result. Each detection result is compared with all the ground truths. The positive prediction results with the corresponding IoU larger than the threshold are considered as true positives (TP); the rest are considered as FPs. In addition, if a ground truth does not have a corresponding detection result, we consider it as a false negative (FN). Then, the precision and recall metrics are formulated as

$$\text{Precision}=\frac{\text{TP}}{\text{TP}+\text{FP}} \tag{12}$$

$$\text{Recall}=\frac{\text{TP}}{\text{TP}+\text{FN}}. \tag{13}$$

The precision–recall curve (PRC) represents the relationship between precision and recall. For a particular detector, the overall trend of PRC is that the recall increases with the

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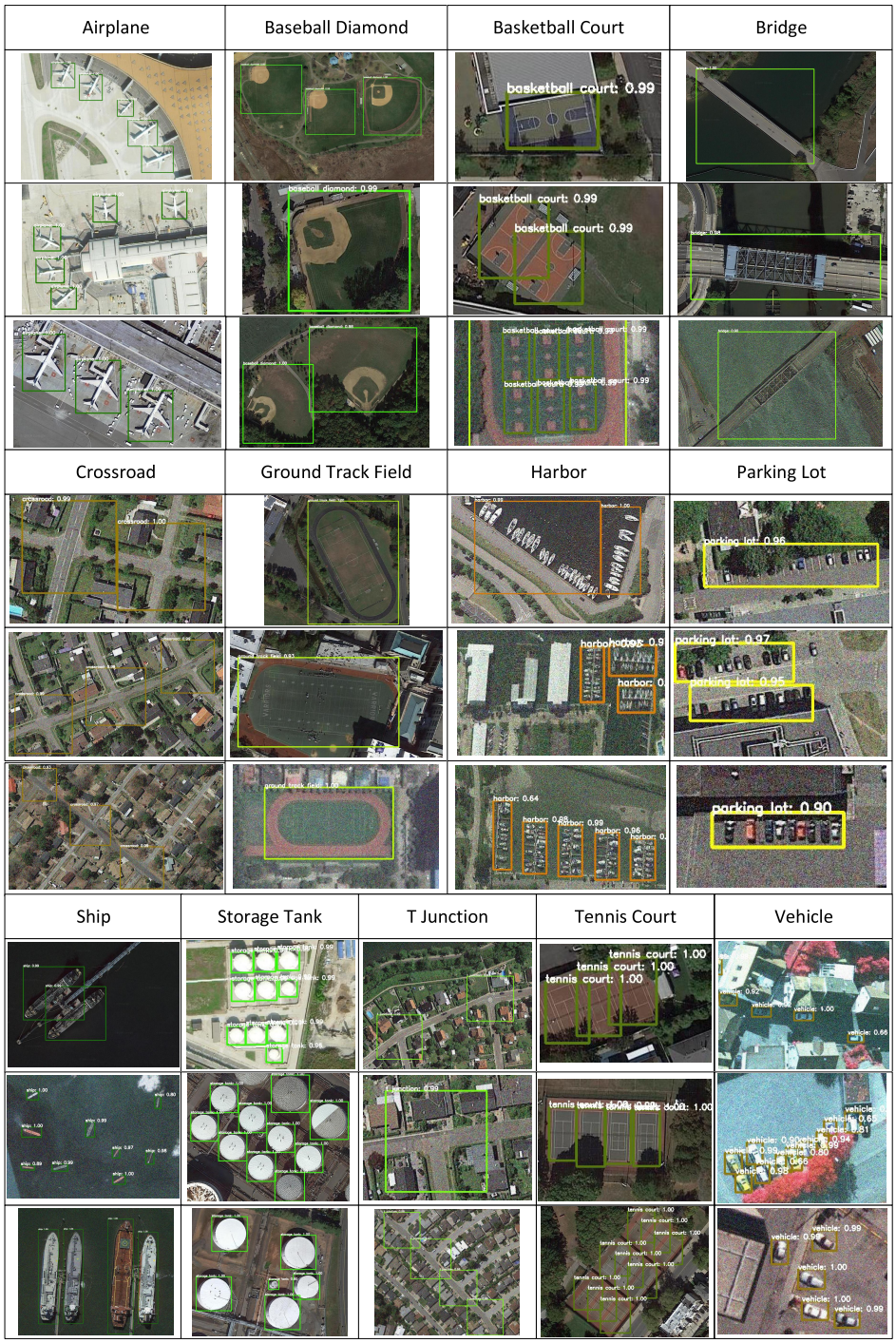


Fig. 6. Detection results on the HRRSD data set.

fall of precision. AP indicates the area under the PRC across the entire recall range. AP is a more comprehensive metric; a higher AP indicates a better detector.

In our report of experimental results, as shown in Tables II–VI, AP represents the average of AP across IoU thresholds from 0.5 to 0.95 with an interval of 0.05, and AP@50, AP@60, AP@70, AP@80, and AP@90 represent the AP with IoU thresholds of 0.5, 0.6, 0.7, 0.8, and 0.9, respectively.

C. Implementation Details

We implemented our network on the open-source Pytorch framework (https://pytorch.org/) and run the program on a 64-bit Ubuntu 14.04 server, which has 110-GB memory, Intel XeonE5-2670 CPU, and GeForce GTX 1080Ti GPU.

During training, we first resized the images to 1333×800 and randomly flipped these resized images with a probability of 0.5. After that, we fed these training samples to the network. We pretrained our ResNet-50 backbone on the ImageNet data

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TABLE II
ABLATION EXPERIMENTS ON HRRSD

	AP	AP@50	AP@60	AP@70	AP@80	AP@90
Baseline	58.4	85.5	83.0	75.1	54.3	15.8
MSRPN	59.1	85.2	83.0	75.7	56.7	17.9
MSRPN+HEMN	62.6	85.9	83.9	78.4	62.4	25.4

TABLE III
ABLATION STUDIES OF MSRPN ON HRRSD

	AP	AP@50	AP@60	AP@70	AP@80	AP@90
Baseline	58.4	85.5	83.0	75.1	54.3	15.8
AFM	58.8	85.2	83.1	75.7	55.7	16.6
AFM+Inception	59.0	85.4	83.2	75.8	55.5	16.4
AFM+LCB	59.1	85.2	83.0	75.7	56.7	17.9

set [51] and fine-tuned it on the train subset of the HRRSD data set. We trained the entire network for 108 000 iterations, with an initial learning rate of 0.002 for the first 80 000 iterations, 0.0002 for next 20 000 iterations, and 0.00002 for the last 8000 iterations. We trained the network using a stochastic gradient descent algorithm, with one image per batch. Weight decay and momentum were set as 0.0001 and 0.9, respectively. We tested our network on the test subset of the HRRSD data set; this procedure was also processed based on the resized images.

TABLE IV
ABLATION STUDIES OF ADAPTIVE WEIGHT ON HRRSD

	AP	AP@50	AP@60	AP@70	AP@80	AP@90
γ =0	62.2	85.8	83.7	78.1	61.4	24.9
γ =0.05	62.6	85.9	83.9	78.4	62.4	25.0
γ =0.1	62.5	85.9	84.0	78.5	62.2	24.9
γ =0.2	62.3	85.9	83.8	78.5	61.6	24.3

D. Experiments and Discussion

We used the HRRSD data set to train, verify, and test the proposed MSHEMN and achieved excellent results in object detection of high-resolution RSIs. Some object detection results of HRRSD are shown in Fig. 6. From these figures, we can see that our proposed network has a good detection performance for objects under complex backgrounds and large-scale range objects.

For convincing evaluation, we use Faster R-CNN+ as baseline, which is based on Faster R-CNN [12]. The backbone feature extraction network is replaced with ResNet-50. RoI align and FPN are also investigated in this model.

The rest of this section includes four parts of experiments. First, we analyze the importance of each component in the proposed architecture by carrying out ablation experiments on the HRRSD data set. Second, we compare our proposed method with existing methods, including state-of-the-art deep learning-based methods, by using AP metrics. Third, we verify the classification improvement by analyzing the classification error rate on HRRSD. Finally, processing time and size of model of our method and other methods for making comparison are reported.

1) Ablation Experiments:

a) Overall ablation studies: In this section, we carried out ablation experiments to verify the effectiveness of the proposed architecture, including HEMN and MSRPN on the HRRSD data set. Parameter γ in adaptive weight was set as 0.05. Our ablation experiment, proceeding from the baseline gradually to all incorporated components, was conducted on the HRRSD data set, and the result is shown in Table II.

It can be seen that we outperform the baseline by 4.2 points on the AP. More concretely, HEMN improves the AP by 3.5 points and MSRPN improves the mAP by 0.7 points.

The comparison of the detection results between our proposed method and the baseline is shown in Fig. 7. We can see that due to the proposed MSRPN and HEMN framework, our method settles the problem of detecting objects under complex background effectively.

b) Ablation studies on MSRPN: We replaced the top–down pathway feature pyramid architecture of Faster R-CNN+ with MSRPN to prove the effectiveness of MSRPN. Comparative experiments with Inception-A block that is used in [52] is also conducted. The architecture of the Inception-A block is shown in Fig. 8, and the experimental results are shown in Table III. AFM and LCB bring 0.4 and 0.3 points higher AP, respectively. To be more specific, most of the improvements are from AP@70 to AP@95. This result validates that MSRPN improves the localization accuracy. The performance of the Inception block is very similar to LCB, which means that more complex module is not useful for further improvement.

c) Ablation studies on adaptive weight: As described in Section III-B, we add an additional adaptive weight to the original cross-entropy loss function of the detection network to make the detection network focus on the hard examples. To prove the effectiveness of the adaptive weight, we train a set of networks with different γ values that range from 0 to 0.2 and then test these networks on the HRRSD data set. The experimental results are shown in Table IV. It can be seen that using an adaptive weight can improve the detection performance. The network achieves the best performance when γ is set to 0.05, and the AP value is improved from 62.2 to 62.6.

2) Comparison With Other Methods on the HRRSD Data Set: To prove the validity of our method, we compared our proposed method with six state-of-the-art methods on the HRRSD data set. They were Faster R-CNN [12], GACL Faster R-CNN [39], Mask R-CNN [14], RetinaNet [38], YOLOv3 [18], and Cascade R-CNN [46]. We conducted this experiment with the help of toolbox MMDetection [53], Faster R-CNN, Mask R-CNN, RetinaNet, and Cascade R-CNN used ResNet-50 as backbone, and YOLOv3 used DarkNet-53

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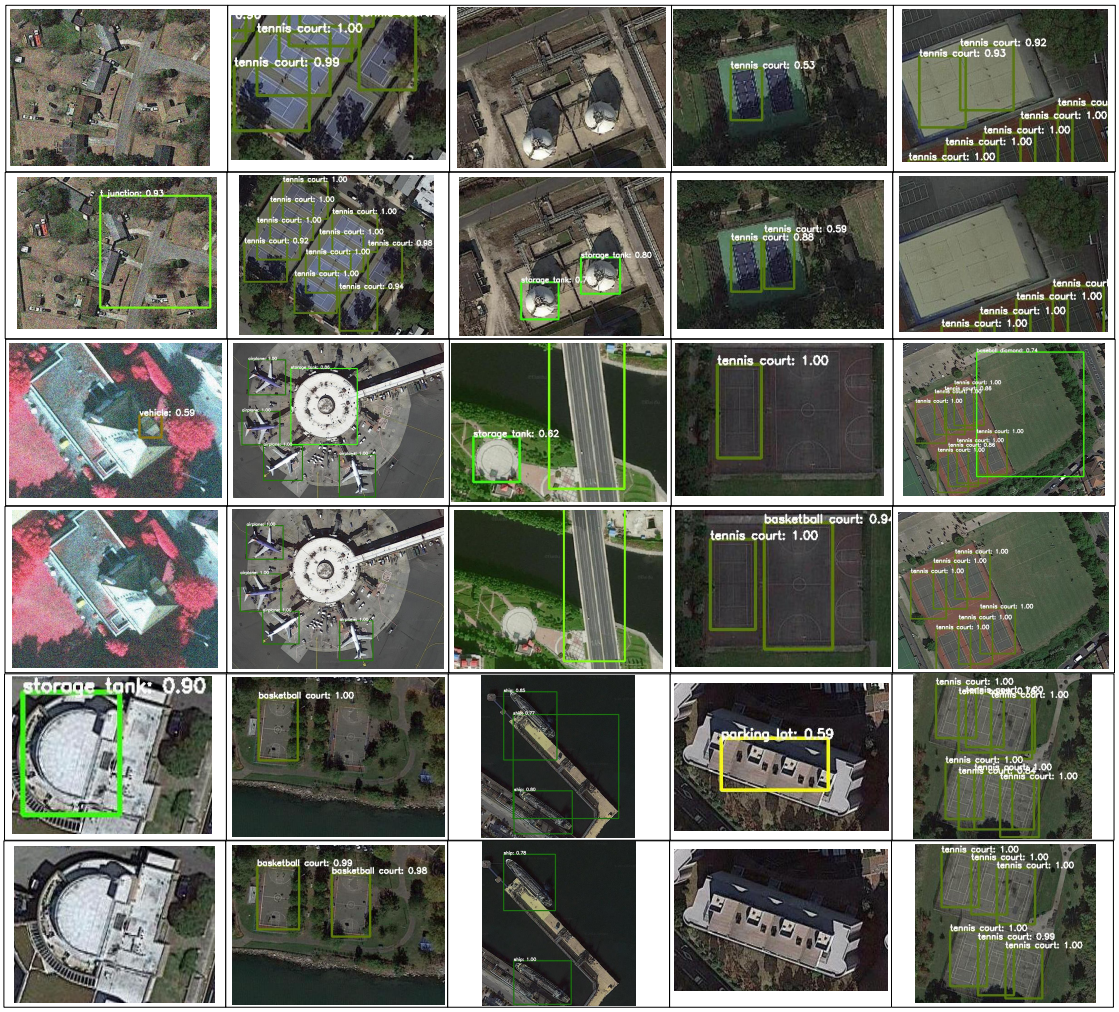


Fig. 7. Comparison of our proposed method and Faster R-CNN with FPN on HRRSD test data set. The first, third, and fifth rows are the detection results of Faster R-CNN with FPN, the second, fourth, and sixth rows are the detection results of our proposed method. We only display the bounding boxes that have a confidence score larger than 0.5 for both cases.

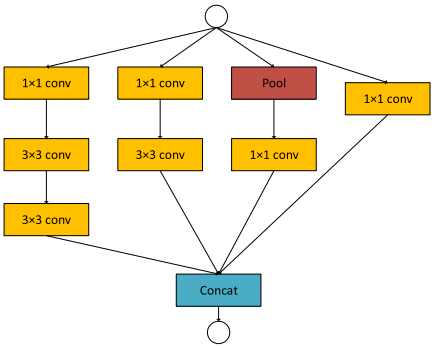


Fig. 8. Architecture of Inception-A block: "conv" represents the convolutional layer, "relu" represents the activation function, "pool" represents the average pooling, and "concat" represents concatenate.

as backbone. For GACL Faster R-CNN, the researchers have reported the AP_{50} results on the HRRSD data set, but no code is shared; therefore, we list the results reported in their publications directly. The results of experiment on the HRRSD data set are shown in Tables V and VI. The detection results in these tables show that our model achieves better performance compared with other methods on the HRRSD data set.

3) Classification Improvement Verification: To prove the classification improvement brought by our proposed method clearly, we replaced the detection network of Faster R-CNN+ with HEMN. After that, we analyzed the percentage of inaccurate labeling of the prediction boxes with different confidence score produced by both networks. The results are shown in Table VII. It can be seen in the table that, after using the HEMN, the classification error rate of prediction boxes is reduced significantly.

4) Speed and Complexity: In our experimental environment, the inference time of our method and Faster R-CNN, Mask R-CNN, RetinaNet, YOLOv3, and Cascade R-CNN is listed in Table VIII. The size of the model is also listed in Table VIII as the metric of complexity.

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TABLE V
COMPARISON OF AP@50 WITH OTHER METHODS ON HRRSD

	YOLOv3	Retina Net	Faster RCNN	GACL Faster RCNN	Mask RCNN	Cascade RCNN	Faster RCNN+	MSHEMN
Airplane	90.7	90.8	90.9	90.8	90.8	90.9	90.9	90.9
Baseball Diamond	76.4	86.8	87.4	87.2	88.7	88.7	88.9	88.5
Basketball Court	66.3	65.7	65.0	49.7	68.3	68.7	68.2	74.0
Bridge	75.2	85.5	84.3	85.6	87.6	88.5	88.7	88.3
Crossroad	75.0	87.4	88.7	88.2	88.8	88.8	88.9	88.7
Ground Track Field	89.4	90.7	90.7	90.7	90.7	90.8	90.8	90.8
Harbor	45.9	89.2	89.1	89.7	89.5	90.0	90.0	88.8
Parking Lot	42.3	60.1	64.1	65.3	66.5	63.0	67.7	68.1
Ship	73.6	88.1	87.5	88.5	89.6	89.8	89.5	89.6
Storage Tank	77.0	89.9	89.8	89.2	89.9	90.0	89.9	89.9
T Junction	59.4	70.8	79.5	75.0	79.1	74.3	77.8	78.7
Tennis Court	89.2	89.9	89.8	80.8	90.3	90.3	90.2	90.4
Vehicle	73.3	89.4	89.6	86.9	90.1	90.3	90.1	90.2
Mean AP@50	71.8	83.4	84.3	82.1	85.4	84.9	85.5	**85.9**

TABLE VI
COMPARISON OF AP WITH OTHER METHODS ON HRRSD

	AP	AP@50	AP@60	AP@70	AP@80	AP@90
YOLOv3	38.8	71.8	63.6	48.0	24.4	4.0
Retina Net	57.8	83.4	80.6	72.5	54.5	19.5
Faster RCNN	56.6	84.3	82.0	72.1	50.7	13.7
Mask RCNN	58.7	85.4	83.0	74.8	53.7	17.0
Cascade RCNN	61.4	84.9	83.3	76.5	60.4	24.2
Faster RCNN+	58.4	85.5	83.0	75.1	54.3	15.8
MSHEMN	**62.6**	85.9	83.9	78.4	62.4	25.4

TABLE VII
COMPARISON OF CLASSIFICATION ERROR RATE ON HRRSD

confidence	⩾0.1	⩾0.5	⩾0.75	⩾0.9
Faster RCNN+	30.7%	16.4%	11.6%	8.1%
HEMN	26.8%	13.9%	9.7%	6.5%

TABLE VIII
COMPARISON OF SPEED AND COMPLEXITY WITH OTHER METHODS

	inference time(ms)	model size(MB)
YOLOv3	36.0	247
Retina Net	67.2	278
Faster RCNN	212.8	265
Mask RCNN	81.9	331
Cascade RCNN	112.3	553
Faster RCNN+	75.6	315
MSHEMN	120.9	552

V. CONCLUSION

In this article, an effective deep learning-based object detection framework for high-resolution RSIs named MSHEMN is proposed. This framework provides a solution to the problem of detecting the large-scale range objects and objects under complex backgrounds in RSIs. For the first problem, we propose MSRPN to extract and fuse multiscale features. For the second problem, we propose a cascade multistage detection network HEMN to make the network focus on the hard examples during training by changing the distribution of input data for each stage. The experiment on the HRRSD data set shows that our framework improves the AP by 4.2 points compared with the backbone Faster R-CNN+ and outperforms the state-of-the-art framework. The reduction of classification error rate also demonstrates that our framework can reduce classification errors significantly, thus proving the effectiveness of our framework. We are going to extend our work in several directions in the future. First, we will implement our model on RSIs that are under more complex situation, for example, challenging weather conditions such as fog, to verify the effectiveness of our model; second, to further overcome the size and distribution imbalance in RSIs and increase the detection performance, we are going to further modify the existing proposal sampling strategy in the training phase; third, we are going to further modify the loss function to balance the classification accurate and localization accurate.

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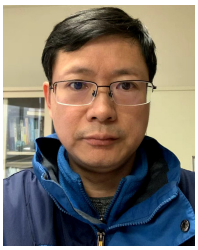
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